

01 MOTIVATION

Flood prediction systems support disaster preparedness and emergency response. However, AI-only predictions may produce incorrect decisions in uncertain situations. Human-in-the-Loop (HITL) AI introduces targeted human review to improve prediction reliability while maintaining efficient automated decision support.

Goal

Develop a Human-in-the-Loop flood prediction framework that:

- Improves prediction reliability
- Reduces critical errors
- Provides explainable decisions
- Supports trustworthy disaster-risk management

DATASETS



Flood Risk Dataset

Dataset Characteristics

- Environmental variables
- Hydrological indicators
- Infrastructure factors
- Flood Probability
- Flood Risk Label

Key Features

Environmental Factors

- Watersheds
- Monsoon Intensity
- Climate Change
- Dam Quality
- Sitation
- Disaster Preparedness

METHODOLOGY WORKFLOW

1

Data Collection

Load flood prediction dataset.

2

Data Preprocessing

Handle missing values and clean data.

3

Feature Engineering

Create Flood Risk label and prepare features.

4

Model Training

Train Random Forest Classifier.

5

AI Prediction

Generate flood risk predictions.

6

Human Review

Review uncertain prediction.

7

Threshold Analysis

Evaluate confidence thresholds.

8

Error Analysis

Analyze False Positives and False Negatives.

9

Explainability Analysis

SHAP feature importance

10

Final Recommendations

Human-in-the-Loop decision support.

RESEARCH QUESTIONS

- RQ1** How accurately can AI-only model predict flood risk?
- RQ2** How much does human review improve flood-risk prediction accuracy?
- RQ3** What are the critical prediction errors made by the AI-only system?
- RQ4** Which environmental and infrastructure factors most influence flood-risk prediction?
- RQ5** What is the trade-off between prediction accuracy and human review workload?

05 KEY RESULTS (HIGHLIGHTS)

Key Findings

RQ1: AI-Only Performance

- Accuracy: 90.8%
- Precision: 91.7%
- Recall: 90.2%
- F1 Score: 90.9%

RQ2: Human Review Impact

- Accuracy improved from **90.8%** → **~100%**
- Human review corrected uncertain predictions

RQ3: Critical Errors

- False Positives: **417**
- False Negatives: **502**

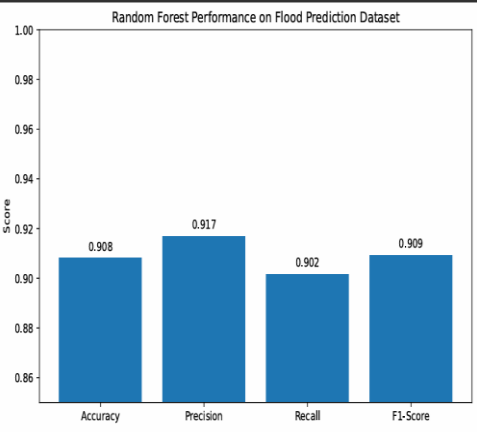
RQ4: Most Important Features

- Watersheds
- Disaster Preparedness
- Siltation
- Dam Quality

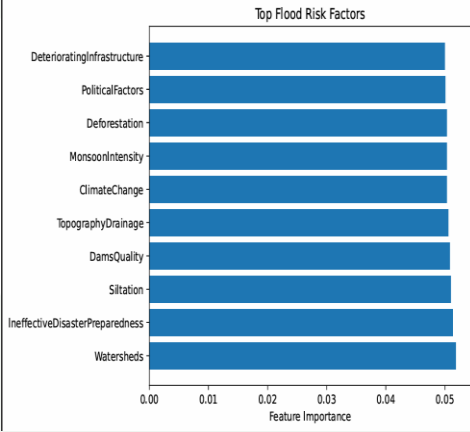
RQ5: Accuracy vs Workload

- Higher review workload yields higher prediction accuracy

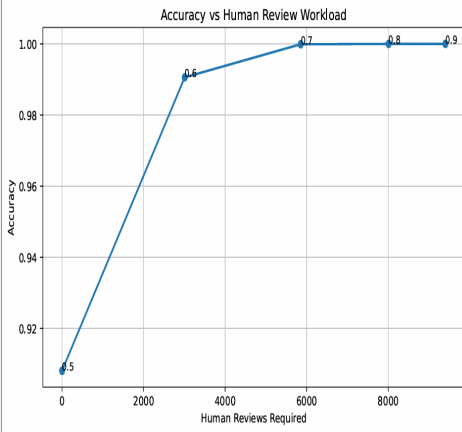
RQ1 – AI-Only Performance Metrics



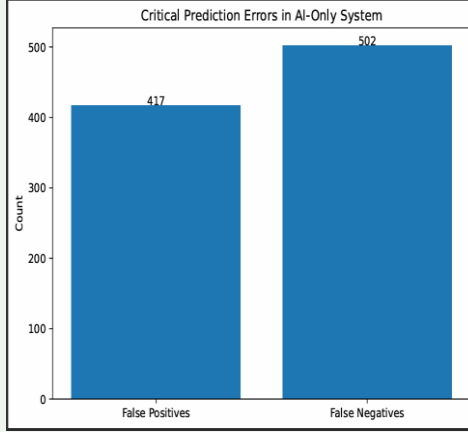
RQ4 – Feature Importance Analysis



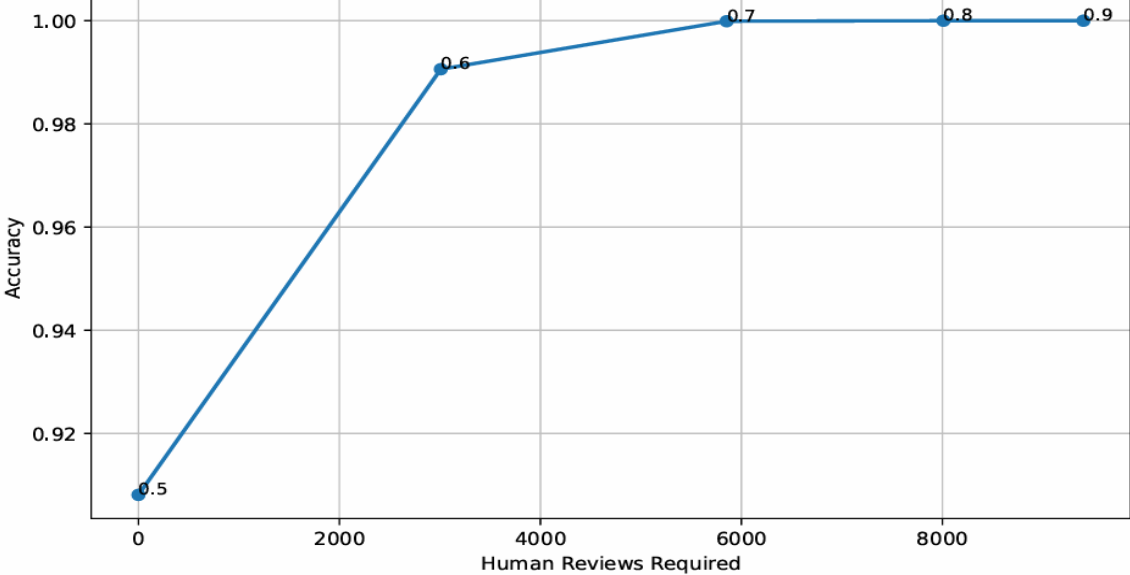
RQ5 – Accuracy vs Human Review Workload



RQ3 – Critical Prediction Errors



Accuracy vs Human Review Workload



CONTRIBUTIONS

- ✓ Human-in-the-Loop flood prediction framework
- ✓ Confidence-based human review
- ✓ Explainable AI using SHAP
- ✓ Critical error reduction analysis
- ✓ Feature importance identification
- ✓ Practical decision support for flood management

08 CONCLUSION

Key Conclusions

- Human review improves flood prediction reliability.
 - SHAP identifies key flood-risk drivers.
 - Critical prediction errors can be analyzed systematically.
 - Accuracy increases with confidence-based review.
 - HITL supports practical flood-risk decision making.
- This makes it easier for examiners to scan.

